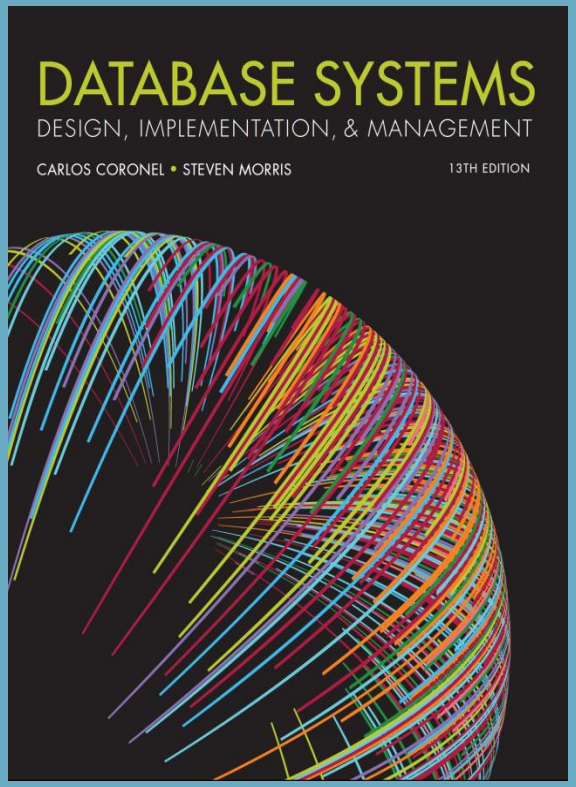


Chapter 6

Normalization of Database Tables

Learning Objectives

- After completing this chapter, you will be able to:
  - Explain normalization and its role in the database design process
  - Identify and describe each of the normal forms: 1NF, 2NF, 3NF, BCNF, and 4NF
  - Explain how normal forms can be transformed from lower normal forms to higher normal forms
  - Apply normalization rules to evaluate and correct table structures
  - Identify situations that require denormalization to generate information efficiently
  - Use a data-modeling checklist to check that the ERD meets a set of minimum requirements

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Normalization

- **Normalization -** Putting data right from the designer's point of view – making it normal by evaluating and correcting table structures to minimize data redundancies.

- Database design is done through 2 approaches:
  1. Top-down approach – first entities and relationships are identified, the ER diagram is made and mapped with the tables.
  2. Bottom – up approach – look at data files and apply relational database concepts to reduce data anomalies.

- Normalization uses the bottom-up approach

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Database Tables and Normalization (1 of 2)

- Normalization: evaluating and correcting table structures to minimize data redundancies
  - Reduces data anomalies
  - Assigns attributes to tables based on determination
- Normal forms
  - First normal form (1NF)
  - Second normal form (2NF)
  - Third normal form (3NF)

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Database Tables and Normalization (2 of 2)

- Structural point of view of normal forms
  - Higher normal forms are better than lower normal forms
  - Properly designed 3NF structures meet the requirement of fourth normal form (4NF)
- Denormalization: produces a lower normal form
  - Results in increased performance and greater data redundancy

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Need for Normalization

- Used while designing a new database structure
  - Analyzes the relationship among the attributes within each entity
  - Determines if the structure can be improved through normalization
  - Improves the existing data structure and creates an appropriate database design

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Normalization Process (1 of 5)

- Objective is to ensure that each table conforms to the concept of well-formed relations
  - Each table represents a single subject
  - Each row/column intersection contains only one value and not a group of values
  - No data item will be unnecessarily stored in more than one table
  - All nonprime attributes in a table are dependent on the primary key
  - Each table has no insertion, update, or deletion anomalies

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Normalization Process (2 of 5)

- Ensures that all tables are in at least 3NF
  - Higher forms are not likely to be encountered in business environment
- Works one relation at a time
  - Identifies the dependencies of a relation (table)
  - Progressively breaks the relation up into a new set of relations

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Normalization Process (3 of 5)

Table 6.2: Normal Forms		
Normal Form	**Characteristic**	**Section**
First normal form (1NF)	Table format, no repeating groups, and PK identified	6-3a
Second normal form (2NF)	1NF and no partial dependencies	6-3b
Third normal form (3NF)	2NF and no transitive dependencies	6-3c
Boyce-Codd normal form (BCNF)	Every determinant is a candidate key (special case of 3NF)	6-6a
Fourth normal form (4NF)	3NF and no independent multivalued dependencies	6-6b

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Normalization Process (4 of 5)

Table 6.3: Functional Dependence Concepts	
Concept	**Definition**
Functional dependence	The attribute B is fully functionally dependent on the attribute A if each value of A determines one and only one value of B. Example: PROJ_NUM S PROJ_NAME (read as PROJ_NUM functionally determines PROJ_NAME) In this case, the attribute PROJ_NUM is known as the determinant attribute, and the attribute PROJ_NAME is known as the dependent attribute.
Functional dependence (generalized definition)	Attribute A determines attribute B (that is, B is functionally dependent on A) if all (generalized definition) of the rows in the table that agree in value for attribute A also agree in value for attribute B.
Fully functional dependence (composite key)	If attribute B is functionally dependent on a composite key A but not on any subset of that composite key, the attribute B is fully functionally dependent on A.

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Normalization Process (5 of 5)

- Partial dependency: functional dependence in which the determinant is only part of the primary key
  - Assumption: one candidate key
  - Straight forward
  - Easy to identify
- Transitive dependency: attribute is dependent on another attribute that is not part of the primary key
  - More difficult to identify among a set of data
  - Occur only when a functional dependence exists among nonprime attributes

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Conversion to First Normal Form (1NF) (1 of 3)

- Repeating group: group of multiple entries of same type can exist for any single key attribute occurrence
  - Reduces data redundancies
- Three step procedure
  - Eliminate the repeating groups
  - Identify the primary key
  - Identify all dependencies
- Dependency diagram: depicts all dependencies found within given table structure
  - Helps to get an overview of all relationships among table's attributes
  - Makes it less likely that an important dependency will be overlooked

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Need for Normalization

- Example: company that manages building projects
  - Charges its clients by billing hours spent on each contract
  - Hourly billing rate is dependent on employee's position
  - Periodically, report is generated that contains information such as displayed

The Need for Normalization

TABLE 6.1 A Sample Report Layout

PROJECT NUMBER	PROJECT NAME	EMPLOYEE NUMBER	EMPLOYEE NAME	JOB CLASS	CHARGE/ HOUR	HOURS BILLED	TOTAL CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	$ 84.50	23.8	$ 2,011.10
		101	John G. News	Database Designer	$105.00	19.4	$ 2,037.00
		105	Alice K. Johnson *	Database Designer	$105.00	35.7	$ 3,748.50
		106	William Smithfield	Programmer	$ 35.75	12.6	$ 450.45
		102	David H. Senior	Systems Analyst	$ 96.75	23.8	$ 2,302.65
				Subtotal			**$10,549.70**
18	Amber Wave	114	Annelise Jones	Applications Designer	$ 48.10	24.6	$ 1,183.26
		118	James J. Frommer	General Support	$ 18.36	45.3	$ 831.71
		104	Anne K. Ramoras *	Systems Analyst	$ 96.75	32.4	$ 3,134.70
		112	Darlene M. Smithson	DSS Analyst	$ 45.95	44.0	$ 2,021.80
				Subtotal			**$ 7,171.47**
22	Rolling Tide	105	Alice K. Johnson	Database Designer	$105.00	64.7	$ 6,793.50
		104	Anne K. Ramoras	Systems Analyst	$ 96.75	48.4	$ 4,682.70
		113	Delbert K. Joenbrood *	Applications Designer	$ 48.10	23.6	$ 1,135.16
		111	Geoff B. Wabash	Clerical Support	$ 26.87	22.0	$ 591.14
		106	William Smithfield	Programmer	$ 35.75	12.8	$ 457.60
				Subtotal			**$13,660.10**
25	Starflight	107	Maria D. Alonzo	Programmer	$ 35.75	24.6	$ 879.45
		115	Travis B. Bawangi	Systems Analyst	$ 96.75	45.8	$ 4,431.15
		101	John G. News *	Database Designer	$105.00	56.3	$ 5,911.50
		114	Annelise Jones	Applications Designer	$ 48.10	33.1	$ 1,592.11
		108	Ralph B. Washington	Systems Analyst	$ 96.75	23.6	$ 2,283.30
		118	James J. Frommer	General Support	$ 18.36	30.5	$ 559.98
		112	Darlene M. Smithson	DSS Analyst	$ 45.95	41.4	$ 1,902.33
				Subtotal			**$17,559.82**
				Total			**$48,941.09**

Note: A * indicates the project leader.

Conversion to First Normal Form (1NF) (2 of 3)

- 1NF describes tabular format in which:
  - All key attributes are defined
  - There are no repeating groups in the table
  - All attributes are dependent on the primary key
- All relational tables satisfy 1NF requirements
- Some tables contain partial dependencies
- Update, insertion, or deletion anomalies
  1. Update Anomalies – Modifying a JOB_CLASS for an employee requires many row alterations
  2. Insertion Anomalies – Just to complete a row definition, a new employee must be assigned to a project.
  3. Deletion Anomalies – Suppose only one employee is associated with the project. Deletion of the employee data will also delete the project data.

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

No Repeating Groups

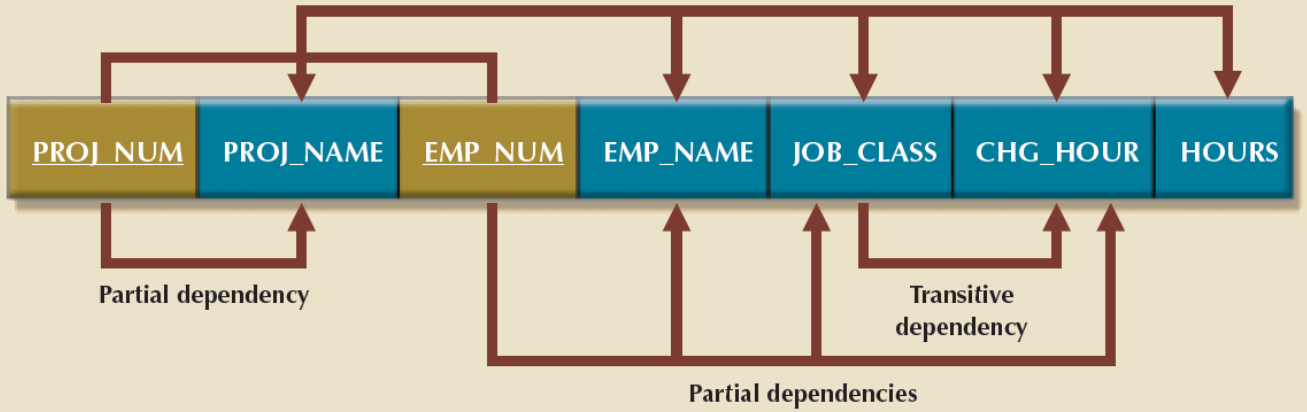
Table name: DATA_ORG_1NF **Database name: Ch06_ConstructCo**

PROJ_NUM	PROJ_NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS
15	Evergreen	103	June E. Arbough	Elect. Engineer	84.50	23.8
15	Evergreen	101	John G. News	Database Designer	105.00	19.4
15	Evergreen	105	Alice K. Johnson *	Database Designer	105.00	35.7
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6
15	Evergreen	102	David H. Senior	Systems Analyst	96.75	23.8
18	Amber Wave	114	Annelise Jones	Applications Designer	48.10	24.6
18	Amber Wave	118	James J. Frommer	General Support	18.36	45.3
18	Amber Wave	104	Anne K. Ramoras *	Systems Analyst	96.75	32.4
18	Amber Wave	112	Darlene M. Smithson	DSS Analyst	45.95	44.0
22	Rolling Tide	105	Alice K. Johnson	Database Designer	105.00	64.7
22	Rolling Tide	104	Anne K. Ramoras	Systems Analyst	96.75	48.4
22	Rolling Tide	113	Delbert K. Joenbrood *	Applications Designer	48.10	23.6
22	Rolling Tide	111	Geoff B. Wabash	Clerical Support	26.87	22.0
22	Rolling Tide	106	William Smithfield	Programmer	35.75	12.8
25	Starflight	107	Maria D. Alonzo	Programmer	35.75	24.6
25	Starflight	115	Travis B. Bawangi	Systems Analyst	96.75	45.8
25	Starflight	101	John G. News *	Database Designer	105.00	56.3
25	Starflight	114	Annelise Jones	Applications Designer	48.10	33.1
25	Starflight	108	Ralph B. Washington	Systems Analyst	96.75	23.6
25	Starflight	118	James J. Frommer	General Support	18.36	30.5
25	Starflight	112	Darlene M. Smithson	DSS Analyst	45.95	41.4

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Conversion to First Normal Form (1NF) (3 of 3)

FIGURE 6.3 FIRST NORMAL FORM (1NF) DEPENDENCY DIAGRAM



1NF (PROJ_NUM, EMP_NUM, PROJ_NAME, EMP_NAME, JOB_CLASS, CHG_HOURS, HOURS)

PARTIAL DEPENDENCIES:
(PROJ_NUM → PROJ_NAME)
(EMP_NUM → EMP_NAME, JOB_CLASS, CHG_HOUR)

TRANSITIVE DEPENDENCY:
(JOB_CLASS → CHG_HOUR)

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

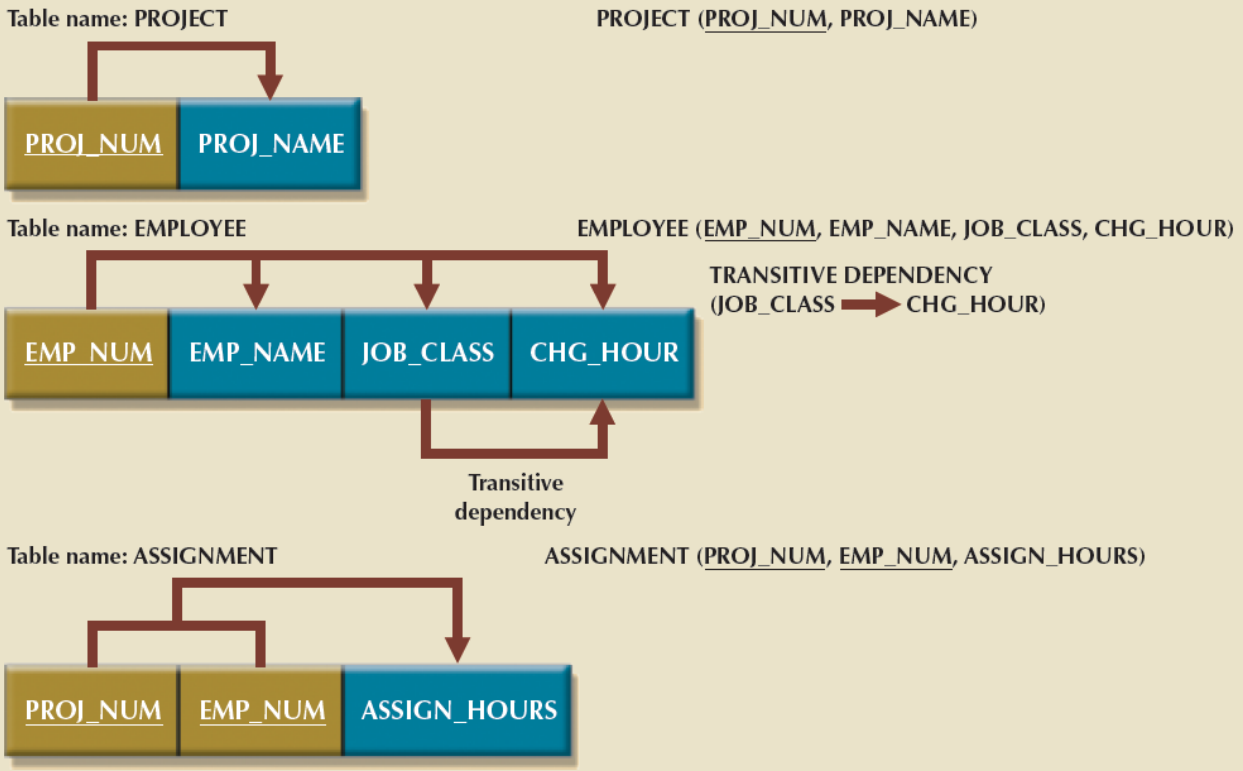
Conversion to Second Normal Form (2NF) (1 of 2)

- Conversion to 2NF occurs only when the 1NF has a composite primary key
  - If the 1NF has a single-attribute primary key, then the table is automatically in 2NF
- The 1NF-to-2NF conversion is simple
  - Make new tables to eliminate partial dependencies
  - Reassign corresponding dependent attributes
- Table is in 2NF when it:
  - Is in 1NF
  - Includes no partial dependencies

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Conversion to Second Normal Form (2NF) (2 of 2)

FIGURE 6.4 SECOND NORMAL FORM (2NF) CONVERSION RESULTS



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Conversion to Third Normal Form (3NF) (1 of 2)

- The data anomalies created by the database organization shown in Figure 6.4 are easily eliminated
  - Make new tables to eliminate transitive dependencies
  - Reassign corresponding dependent attributes
- Table is in 3NF when it:
  - Is in 2NF
  - Contains no transitive dependencies

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Conversion to Third Normal Form (3NF) (2 of 2)

FIGURE 6.5 THIRD NORMAL FORM (3NF) CONVERSION RESULTS

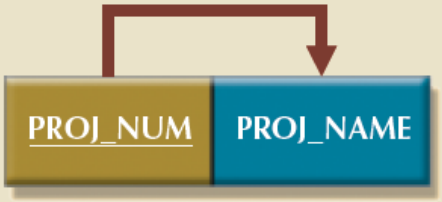


Table name: PROJECT

PROJECT (PROJ_NUM, PROJ_NAME)

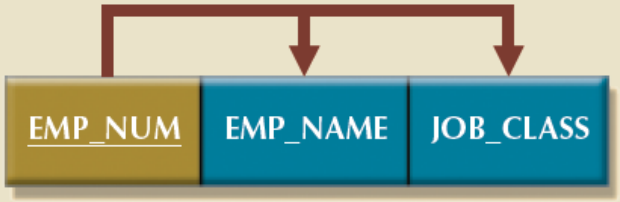


Table name: EMPLOYEE

EMPLOYEE (EMP_NUM, EMP_NAME, JOB_CLASS)



Table name: JOB

JOB (JOB_CLASS, CHG_HOUR)

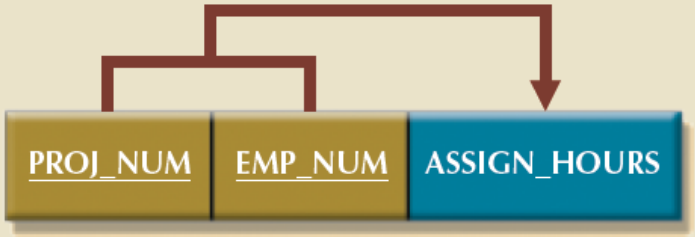


Table name: ASSIGNMENT

ASSIGNMENT (PROJ_NUM, EMP_NUM, ASSIGN_HOURS)

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Improving the Design

- Normalization is valuable because its use helps eliminate data redundancies
  - Evaluate PK assignments and naming conventions
  - Refine attribute atomicity
    - Atomic attribute: cannot be further subdivided
    - Atomicity: characteristic of an atomic attribute
  - Identify new attributes and new relationships
  - Refine primary keys as required for data granularity
    - Granularity: Level of detail represented by the values stored in a table's row
  - Maintain historical accuracy and evaluate using derived attributes

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Completed Database

Table name: PROJECT

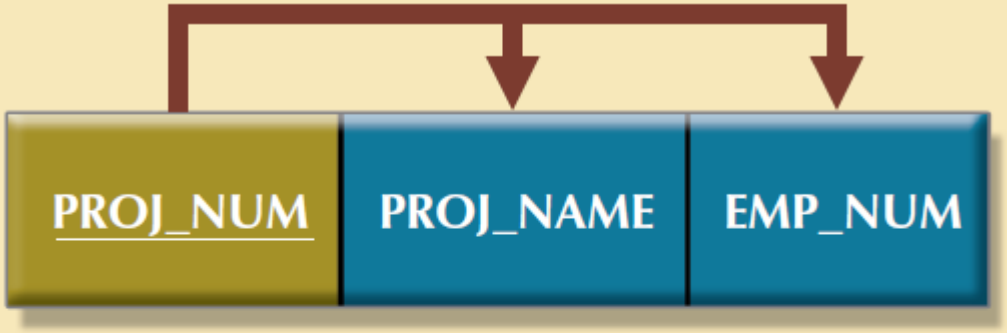


Table name: PROJECT

PROJ_NUM	PROJ_NAME	EMP_NUM
15	Evergreen	105
18	Amber Wave	104
22	Rolling Tide	113
25	Starflight	101

Cengage Learning © 2015

The Completed Database

Table name: JOB Database name: Ch06_ConstructCo



Table name: JOB

JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR
500	Programmer	35.75
501	Systems Analyst	96.75
502	Database Designer	105.00
503	Electrical Engineer	84.50
504	Mechanical Engineer	67.90
505	Civil Engineer	55.78
506	Clerical Support	26.87
507	DSS Analyst	45.95
508	Applications Designer	48.10
509	Bio Technician	34.55
510	General Support	18.36

CENGAGE
Cengage Learning © 2015

The Completed Database

Table name: ASSIGNMENT

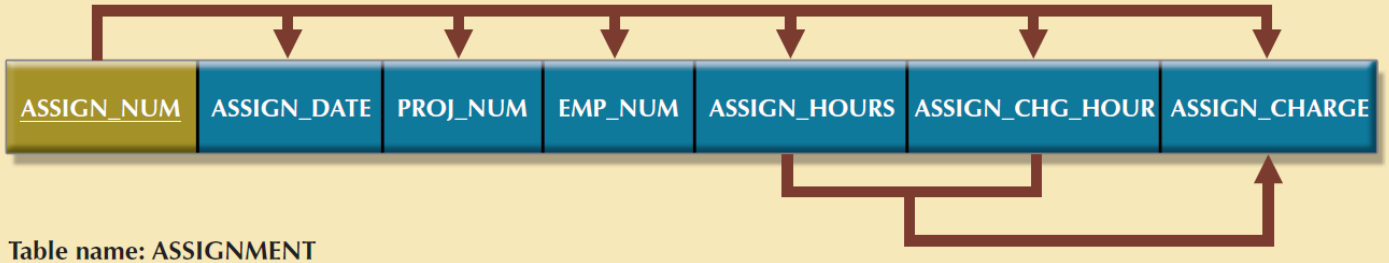


Table name: ASSIGNMENT

ASSIGN_NUM	ASSIGN_DATE	PROJ_NUM	EMP_NUM	ASSIGN_HOURS	ASSIGN_CHG_HOUR	ASSIGN_CHARGE
1001	04-Mar-14	15	103	2.6	84.50	219.70
1002	04-Mar-14	18	118	1.4	18.36	25.70
1003	05-Mar-14	15	101	3.6	105.00	378.00
1004	05-Mar-14	22	113	2.5	48.10	120.25
1005	05-Mar-14	15	103	1.9	84.50	160.55
1006	05-Mar-14	25	115	4.2	96.75	406.35
1007	05-Mar-14	22	105	5.2	105.00	546.00
1008	05-Mar-14	25	101	1.7	105.00	178.50
1009	05-Mar-14	15	105	2.0	105.00	210.00
1010	06-Mar-14	15	102	3.8	96.75	367.65
1011	06-Mar-14	22	104	2.6	96.75	251.55
1012	06-Mar-14	15	101	2.3	105.00	241.50
1013	06-Mar-14	25	114	1.8	48.10	86.58
1014	06-Mar-14	22	111	4.0	26.87	107.48
1015	06-Mar-14	25	114	3.4	48.10	163.54
1016	06-Mar-14	18	112	1.2	45.95	55.14
1017	06-Mar-14	18	118	2.0	18.36	36.72
1018	06-Mar-14	18	104	2.6	96.75	251.55
1019	06-Mar-14	15	103	3.0	84.50	253.50
1020	07-Mar-14	22	105	2.7	105.00	283.50
1021	08-Mar-14	25	108	4.2	96.75	406.35
1022	07-Mar-14	25	114	5.8	48.10	278.98
1023	07-Mar-14	22	106	2.4	35.75	85.80

Cengage Learning © 2015

The Completed Database

Table name: EMPLOYEE **Database name: Ch06_ConstructCo**

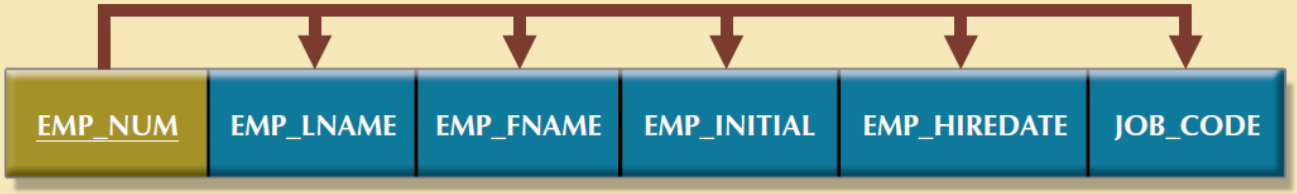
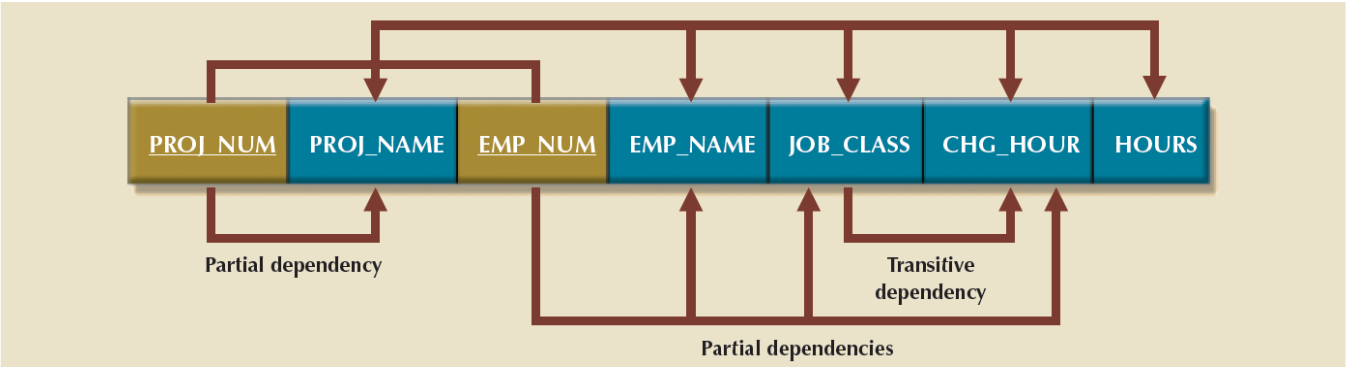


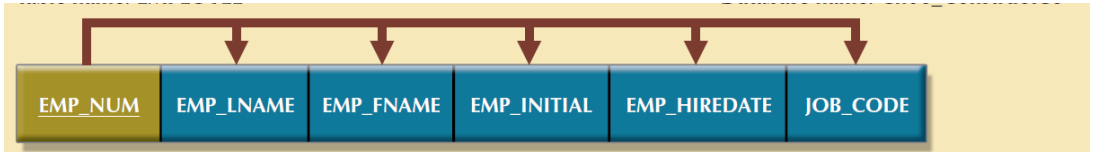
Table name: EMPLOYEE

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE
101	News	John	G	08-Nov-00	502
102	Senior	David	H	12-Jul-89	501
103	Arbough	June	E	01-Dec-97	503
104	Ramoras	Anne	K	15-Nov-88	501
105	Johnson	Alice	K	01-Feb-94	502
106	Smithfield	William		22-Jun-05	500
107	Alonzo	Maria	D	10-Oct-94	500
108	Washington	Ralph	B	22-Aug-89	501
109	Smith	Larry	W	18-Jul-99	501
110	Olenko	Gerald	A	11-Dec-96	505
111	Wabash	Geoff	B	04-Apr-89	506
112	Smithson	Darlene	M	23-Oct-95	507
113	Joenbrood	Delbert	K	15-Nov-94	508
114	Jones	Annelise		20-Aug-91	508
115	Bawangi	Travis	B	25-Jan-90	501
116	Pratt	Gerald	L	05-Mar-95	510
117	Williamson	Angie	H	19-Jun-94	509
118	Frommer	James	J	04-Jan-06	510

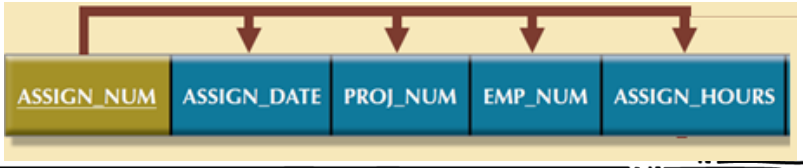
Cengage Learning © 2015











CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Surrogate Key Considerations

- Used by designers when the primary key is considered to be unsuitable
  - System-defined attribute
  - Created an managed via the DBMS
  - Have a numeric value which is automatically incremented for each new row

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Boyce-Codd Normal Form (1 of 4)

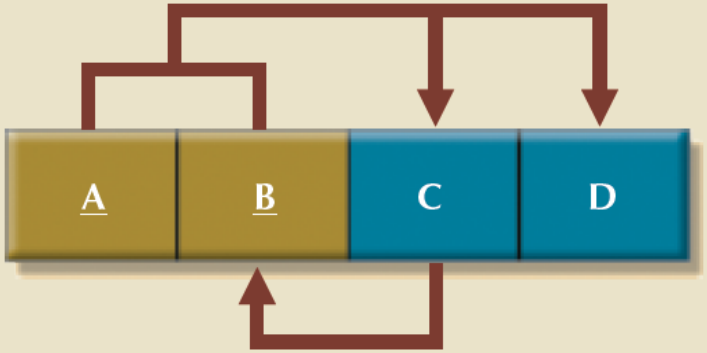
- Every determinant in the table should be a candidate key
  - Candidate key: same characteristics as primary key but not chosen to be the primary key
  - Equivalent to 3NF when the table contains only one candidate key
  - Violated only when the table contains more than one candidate key
  - Considered to be a special case of 3NF

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Boyce-Codd Normal Form (2 of 4)

FIGURE 6.8 A TABLE THAT IS IN 3NF BUT NOT IN BCNF



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Boyce-Codd Normal Form (3 of 4)

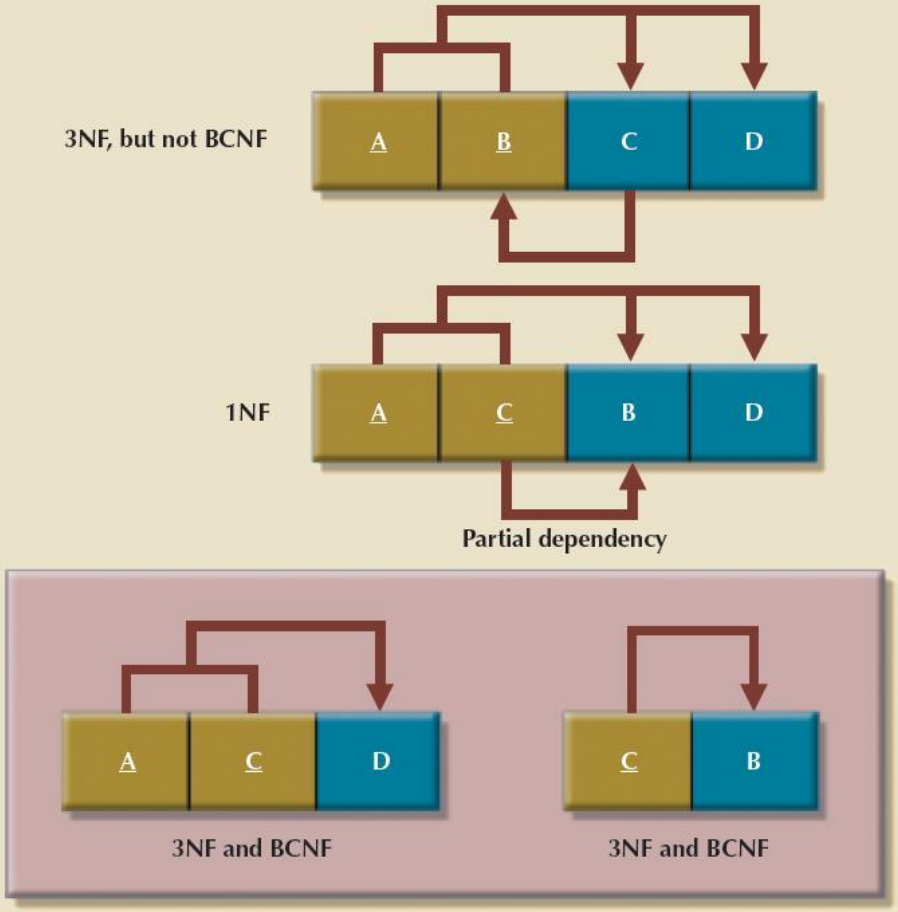
Table 6.5: Sample Data for a BCNF Conversion			
STU_ID	**STAFF_ID**	**CLASS_CODE**	**ENROLL_GRADE**
125	25	21334	A
125	20	32456	C
135	20	28458	B
144	25	27563	C
144	20	32456	B



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

The Boyce-Codd Normal Form (4 of 4)

FIGURE 6.9 DECOMPOSITION TO BCNF



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Fourth Normal Form (4NF) (1 of 2)

- Rules
  - All attributes must be dependent on the primary key, but they must be independent of each other
  - No row may contain two or more multivalued facts about an entity
- Table is in 4NF when it:
  - Is in 3NF
  - Has no multivalued dependencies

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Fourth Normal Form (4NF) (2 of 2)

FIGURE 6.11 TABLES WITH MULTIVALUED DEPENDENCIES

Database name: Ch06_Service

Table name: VOLUNTEER_V1

EMP_NUM	ORG_CODE	ASSIGN_NUM
10123	RC	1
10123	UW	3
10123		4

Table name: VOLUNTEER_V2

EMP_NUM	ORG_CODE	ASSIGN_NUM
10123	RC	
10123	UW	
10123		1
10123		3
10123		4

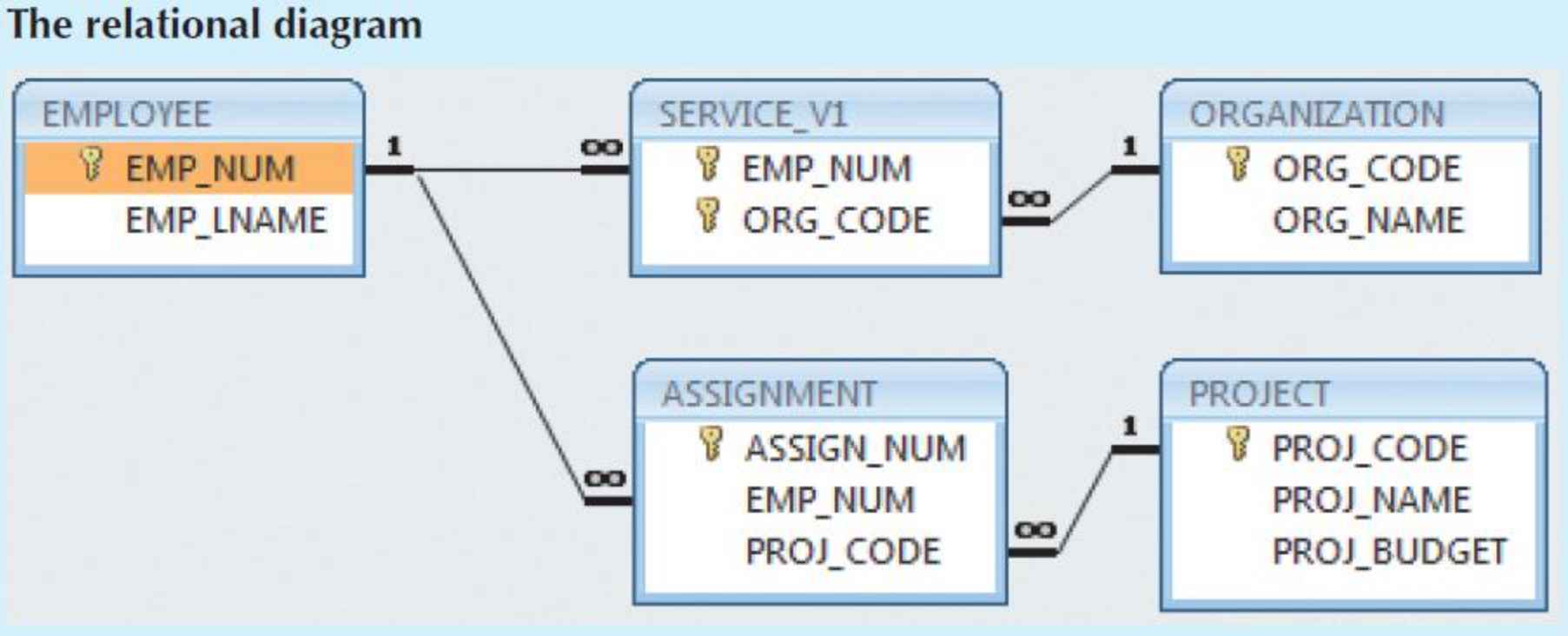
Table name: VOLUNTEER_V3

EMP_NUM	ORG_CODE	ASSIGN_NUM
10123	RC	1
10123	RC	3
10123	UW	4

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

A Set of Tables in 4NF



Cengage Learning © 2015

A Set of Tables in 4NF

Database name: CH06_Service

Table name: PROJECT

PROJ_CODE	PROJ_NAME	PROJ_BUDGET
1	BeThere	1023245.00
2	BlueMoon	20198608.00
3	GreenThumb	3234456.00
4	GoFast	5674000.00
5	GoSlow	1002500.00

Table name: ASSIGNMENT

ASSIGN_NUM	EMP_NUM	PROJ_CODE
1	10123	1
2	10121	2
3	10123	3
4	10123	4
5	10121	1
6	10124	2
7	10124	3
8	10124	5

Table name: EMPLOYEE

EMP_NUM	EMP_LNAME
10121	Rogers
10122	O'Leery
10123	Panera
10124	Johnson

Table name: ORGANIZATION

ORG_CODE	ORG_NAME
RC	Red Cross
UW	United Way
WF	Wildlife Fund

Table name: SERVICE_V1

EMP_NUM	ORG_CODE
10123	RC
10123	UW
10123	WF

Cengage Learning © 2015

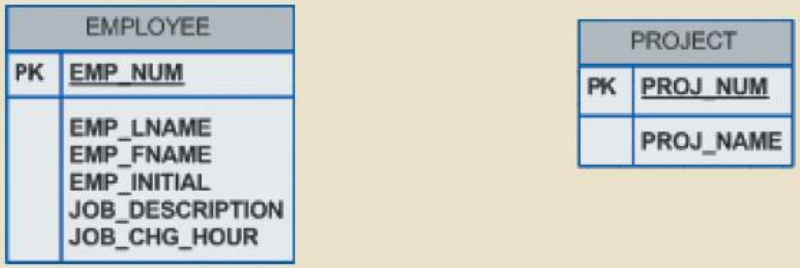
Normalization and Database Design (1 of 6)

- Normalization should be part of the design process
  - Proposed entities must meet required the normal form before table structures are created
- Principles and normalization procedures to be understood to redesign and modify databases
  - ERD is created through an iterative process
  - Normalization focuses on the characteristics of specific entities

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Normalization and Database Design (2 of 6)

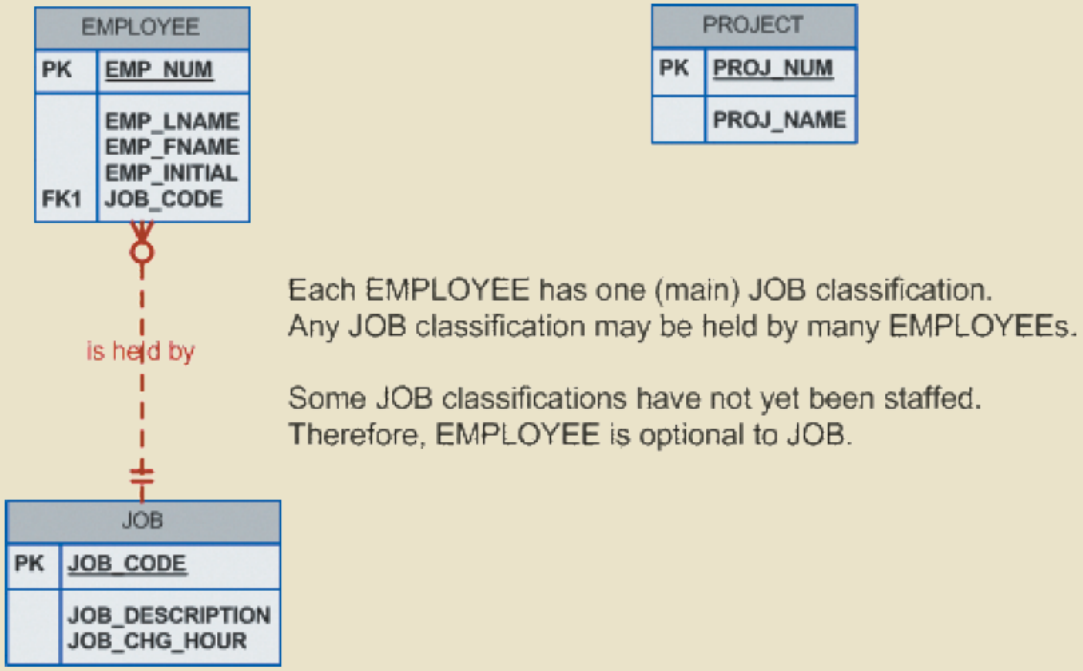
FIGURE 6.13 INITIAL CONTRACTING COMPANY ERD



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Normalization and Database Design (3 of 6)

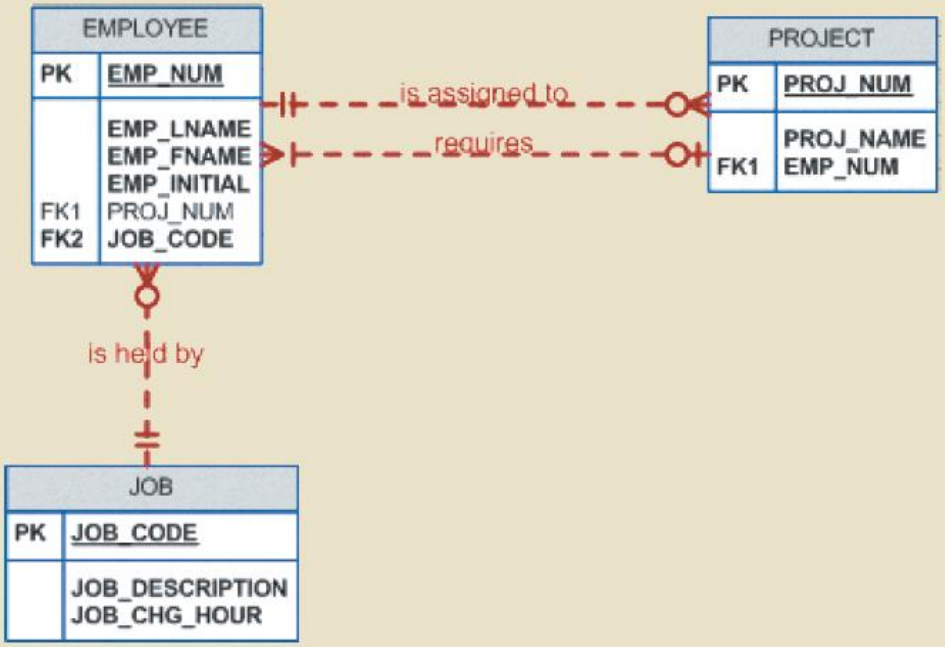
FIGURE 6.14 MODIFIED CONTRACTING COMPANY ERD



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Normalization and Database Design (4 of 6)

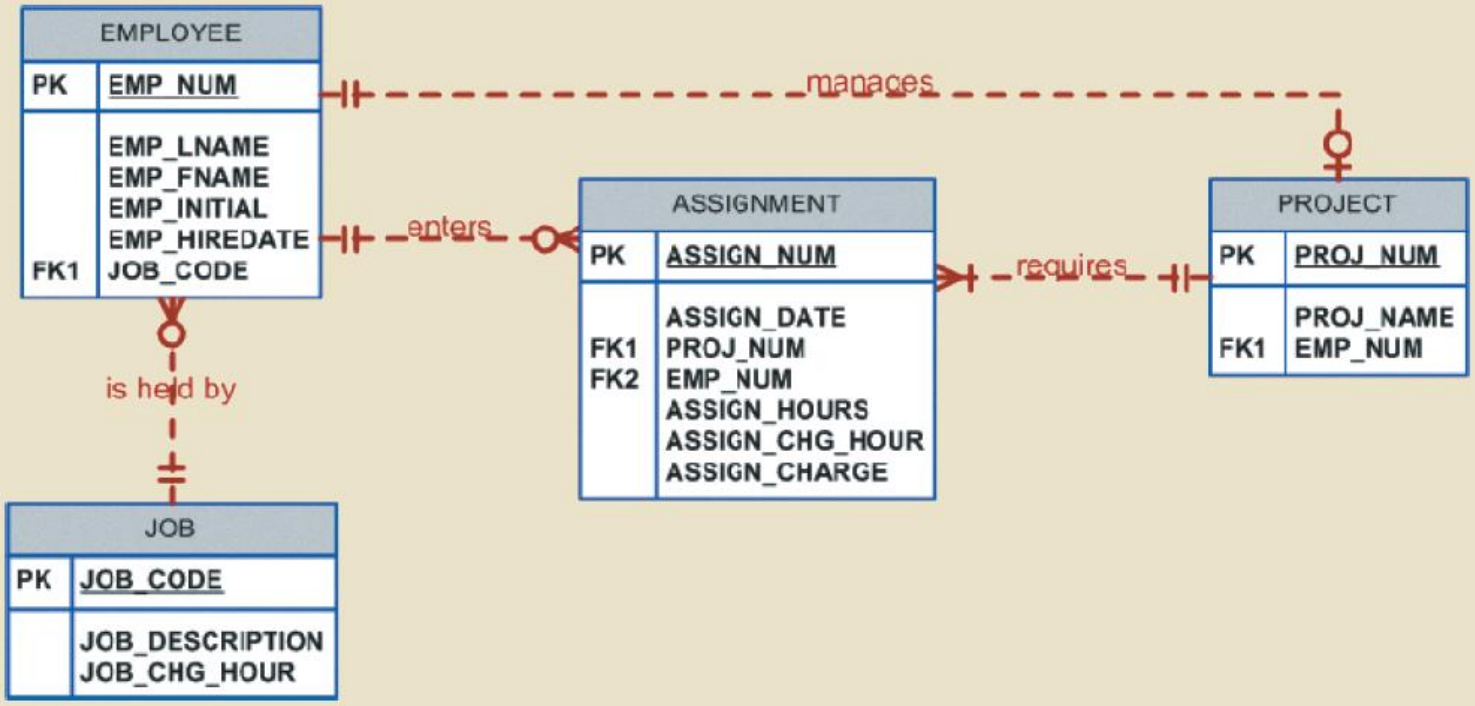
FIGURE 6.15 INCORRECT M:N RELATIONSHIP REPRESENTATION



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Normalization and Database Design (5 of 6)

FIGURE 6.16 FINAL CONTRACTING COMPANY ERD



© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Normalization and Database Design (6 of 6)

FIGURE 6.17 THE IMPLEMENTED DATABASE

Table name: EMPLOYEE

Database name: Ch06_ConstructCo

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_HIREDATE	JOB_CODE
101	News	John	G	08-Nov-00	502
102	Senior	David	H	12-Jul-89	501
103	Arbough	June	E	01-Dec-97	503
104	Ramoras	Anne	K	15-Nov-88	501
105	Johnson	Alice	K	01-Feb-94	502
106	Smithfield	William		22-Jun-05	500
107	Alonzo	Maria	D	10-Oct-94	500
108	Washington	Ralph	B	22-Aug-89	501
109	Smith	Larry	W	18-Jul-99	501
110	Olenko	Gerald	A	11-Dec-96	505
111	Wabash	Geoff	B	04-Apr-89	506
112	Smithson	Darlene	M	23-Oct-95	507
113	Joenbrood	Delbert	K	15-Nov-94	508
114	Jones	Annelise		20-Aug-91	508
115	Bawangi	Travis	B	25-Jan-90	501
116	Pratt	Gerald	L	05-Mar-95	510
117	Williamson	Angie	H	19-Jun-94	509
118	Frommer	James	J	04-Jan-06	510

Table name: JOB

JOB_CODE	JOB_DESCRIPTION	JOB_CHG_HOUR
500	Programmer	35.75
501	Systems Analyst	96.75
502	Database Designer	105.00
503	Electrical Engineer	84.50
504	Mechanical Engineer	67.90
505	Civil Engineer	55.78
506	Clerical Support	26.87
507	DSS Analyst	45.95
508	Applications Designer	48.10
509	Bio Technician	34.55
510	General Support	18.36

Table name: PROJECT

PROJ_NUM	PROJ_NAME	EMP_NUM
15	Evergreen	105
18	Amber Wave	104
22	Rolling Tide	113
25	Starflight	101

Table name: ASSIGNMENT

ASSIGN_NUM	ASSIGN_DATE	PROJ_NUM	EMP_NUM	ASSIGN_HOURS	ASSIGN_CHG_HOUR	ASSIGN_CHARGE
1001	04-Mar-16	15	103	2.6	84.50	219.70
1002	04-Mar-16	18	118	1.4	18.36	25.70
1003	05-Mar-16	15	101	3.6	105.00	378.00
1004	05-Mar-16	22	113	2.5	48.10	120.25
1005	05-Mar-16	15	103	1.9	84.50	160.55
1006	05-Mar-16	25	115	4.2	96.75	406.35
1007	05-Mar-16	22	105	5.2	105.00	546.00
1008	05-Mar-16	25	101	1.7	105.00	178.50
1009	05-Mar-16	15	105	2.0	105.00	210.00
1010	06-Mar-16	15	102	3.8	96.75	367.65
1011	06-Mar-16	22	104	2.6	96.75	251.55
1012	06-Mar-16	15	101	2.3	105.00	241.50
1013	06-Mar-16	25	114	1.8	48.10	86.58
1014	06-Mar-16	22	111	4.0	26.87	107.48
1015	06-Mar-16	25	114	3.4	48.10	163.54
1016	06-Mar-16	18	112	1.2	45.95	55.14
1017	06-Mar-16	18	118	2.0	18.36	36.72
1018	06-Mar-16	18	104	2.6	96.75	251.55
1019	06-Mar-16	15	103	3.0	84.50	253.50
1020	07-Mar-16	22	105	2.7	105.00	283.50
1021	08-Mar-16	25	108	4.2	96.75	406.35
1022	07-Mar-16	25	114	5.8	48.10	278.98
1023	07-Mar-16	22	106	2.4	35.75	85.80

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Denormalization (1 of 2)

- Design goals
  - Creation of normalized relations
  - Processing requirements and speed
- Number of database tables expands
  - Tables are decomposed to conform to normalization requirements
- Joining a larger number of tables
  - Takes additional input/output (I/O) operations and processing logic
  - Reduces system speed
- Defects in unnormalized tables
  - Data updates are less efficient because tables are larger
  - Indexing is more cumbersome
  - No simple strategies for creating virtual tables known as views

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Denormalization (2 of 2)

Table 6.6: Common Denormalization Examples		
Case	**Example**	**Rationale and Controls**
Redundant data	Storing ZIP and CITY attributes in the AGENT table when ZIP determines CITY (see Figure 2.2)	Avoid extra join operations Program can validate city (drop-down box) based on the zip code
Derived data	Storing STU_HRS and STU_CLASS (student classification) when STU_HRS determines STU_CLASS (see Figure 3.28)	Avoid extra join operations Program can validate classification (lookup) based on the student hours
Preaggregated data (also derived data)	Storing the student grade point average (STU_GPA) aggregate value in the STUDENT table when this can be calculated from the ENROLL and COURSE tables (see Figure 3.28)	Avoid extra join operations Program computes the GPA every time a grade is entered or updated STU_GPA can be updated only via administrative routine
Information requirements	Using a temporary denormalized table to hold report data; this is required when creating a tabular report in which the columns represent data that are stored in the table as rows (see Figures 6.17 and 6.18)	Impossible to generate the data required by the report using plain SQL No need to maintain table Temporary table is deleted once report is done Processing speed is not an issue

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Denormalization

FIGURE 6.18 The EVALDATA and FACHIST tables

Table name: EVALDATA

ID	INSTRUCTOR	DEPARTMENT	MEAN	SEMESTER
3730	Alton	INFS	2.2	2002F
3764	Alton	INFS	2.69	2002S
3975	Alton	INFS	2.51	2003F
4172	Alton	INFS	2.13	2003S
4450	Alton	INFS	2.84	2004F
4323	Alton	INFS	2.55	2004S
4571	Alton	INFS	2.91	2005S
2091	Ames	INFS	2.84	1996F
2215	Ames	INFS	2.76	1996S
2332	Ames	INFS	2.79	1997F
2450	Ames	INFS	2.74	1997S
2573	Ames	INFS	2.6	1998F
2699	Ames	INFS	2.92	1998S
2823	Ames	INFS	2.09	1999F
2948	Ames	INFS	2.98	1999S
3071	Ames	INFS	3.35	2000F
3205	Ames	INFS	2.92	2000S
3337	Ames	INFS	3.24	2001F
3473	Ames	INFS	2.79	2001S
3668	Ames	INFS	3.28	2002F
3765	Ames	INFS	3.24	2002S
3976	Ames	INFS	3.19	2003F
4155	Ames	INFS	2.59	2003S
4414	Ames	INFS	3.26	2004F
4269	Ames	INFS	3.31	2004S
4572	Ames	INFS	3.24	2005S
3613	Crandon	INFS	3.79	2002F
3977	Crandon	INFS	3.88	2003F
4040	Crandon	INFS	3.79	2003S
4346	Crandon	INFS	3.95	2004F
4199	Crandon	INFS	3.91	2004S
4573	Crandon	INFS	3.93	2005S

Table name: FACHIST **Database name: Ch06_EVAL**

ID	INSTRUCTOR	DEPARTMENT	LAST1SEM	LAST1MEAN	LAST2SEM	LAST2MEAN	LAST3SEM	LAST3MEAN	LAST4SEM	LAST4MEAN
59602	Alton	INFS	2005S	2.91	2004F	2.84	2004S	2.55	2003F	2.51
59603	Ames	INFS	2005S	3.24	2004F	3.26	2004S	3.31	2003F	3.19
59605	Crandon	INFS	2005S	3.93	2004F	3.95	2004S	3.91	2003F	3.88
59607	Dumas	MGMT	2004F	3.66	2004S	3.69	2003F	3.56	2003S	3.72
59608	Landon	BMOM	2005S	3.57	2004F	3.64	2004S	3.39	2003F	3.57
59610	Lohar	ECON	1999F	3.53	1998F	3.53				
59611	Rolman	INFS	1996S	3.5						

Denormalized

Repeating Group

Normalized

SOURCE: Course Technology/Cengage Learning

Transposing of multirow data (Normalized) to multicolumn data (Denormalized) is complex

Data-Modeling Checklist (1 of 6)

- Business rules
  - Properly document and verify all business rules with the end users
  - Ensure that all business rules are written precisely, clearly, and simply
    - The business rules must help identify entities, attributes, relationships, and constraints
  - Identify the source of all business rules, and ensure that each business rule is justified, dated, and signed off by an approving authority

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Data-Modeling Checklist (2 of 6)

- Data modeling
  - Naming conventions: all names should be limited in length (database-dependent size)
- Entity names:
  - Should be nouns that are familiar to business and should be short and meaningful
  - Should document abbreviations, synonyms, and aliases for each entity
  - Should be unique within the model
  - For composite entities, may include a combination of abbreviated names of the entities linked through the composite entity

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Data-Modeling Checklist (3 of 6)

- Attribute names:
  - Should be unique within the entity
  - Should use the entity abbreviation as a prefix
  - Should be descriptive of the characteristic
  - Should use suffixes such as _ID, _NUM, or _CODE for the PK attribute
  - Should not be a reserved word
  - Should not contain spaces or special characters such as @, !, or &
- Relationship names:
  - Should be active or passive verbs that clearly indicate the nature of the relationship

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Data-Modeling Checklist (4 of 6)

- Entities:
  - Each entity should represent a single subject
  - Each entity should represent a set of distinguishable entity instances
  - All entities should be in 3NF or higher
    - Any entities below 3NF should be justified
  - Granularity of the entity instance should be clearly defined
  - PK should be clearly defined and support the selected data granularity

CENGAGE

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Data-Modeling Checklist (5 of 6)

- Attributes:
  - Should be simple and single-valued (atomic data)
  - Should document default values, constraints, synonyms, and aliases
  - Derived attributes should be clearly identified and include source(s)
  - Should not be redundant unless this is required for transaction accuracy, performance, or maintaining a history
  - Nonkey attributes must be fully dependent on the PK attribute
- Relationships:
  - Should clearly identify relationship participants
  - Should clearly define participation, connectivity, and document cardinality

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Data-Modeling Checklist (6 of 6)

- ER model:
  - Should be validated against expected processes: inserts, updates, and deletions
  - Should evaluate where, when, and how to maintain a history
  - Should not contain redundant relationships except as required (see attributes)
  - Should minimize data redundancy to ensure single-place updates
  - Should conform to the minimal data rule: All that is needed is there, and all that is there is needed

CENGAGE
© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Summary (1 of 2)

- Normalization is a technique used to design tables in which data redundancies are minimized
- A table is in 1NF when all key attributes are defined and all remaining attributes are dependent on the primary key
- A table is in 2NF when it is in 1NF and contains no partial dependencies
- A table is in 3NF when it is in 2NF and contains no transitive dependencies
- A table that is not in 3NF may be split into new tables until all of the tables meet the 3NF requirements
- Normalization is an important part—but only a part—of the design process
- A table in 3NF might contain multivalued dependencies that produce either numerous null values or redundant data

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.

Summary (2 of 2)

- The larger the number of tables, the more additional I/O operations and processing logic you need to join them
- The data-modeling checklist provides a way for the designer to check that the ERD meets a set of minimum requirements

© 2019 Cengage. May not be copied, scanned, or duplicated, in whole or in part, except for use as permitted in a license distributed with a certain product or service or otherwise on a password-protected website for classroom use.



Datatypes

- A datatype identifies the type of data Oracle 12g is expected to store in a column

```
CREATE TABLE tablename(
columnname1 datatype [Default value],
columnname2 datatype [Default value]
);
```

- Example

```
CREATE TABLE Employee(
Emp_Code NUMBER constraint PK_EMP_CODE Primary Key,
Emp_Name VARCHAR2 (30),
Emp_DOB DATE,
Emp_Joiningdate DATE DEFAULT SYSDATE
);
```

Datatypes

Datatype	Description
VARCHAR2(n)	Variable-length character data, where n represents the column's maximum length (max size = 4000 characters) VARCHAR2(9) can contain up to nine letters, numbers or symbols
CHAR(n)	Fixed-length character data, where n represents the column's maximum length. Default size is 1, and the max is 2000 CHAR(9) can contain a maximum of letters, numbers or symbols
NUMBER(p,s)	Numeric column. The p indicates precision and s indicates scale NUMBER(7,2) can store a value up to 99999.99. If precision or scale is not specifies, the column defaults to a precision of 38 digits
DATE	Stores data between January 1, 4712 BC and December 31, 9999 AD. The Date is displayed in the format DD_MON_YY. Other aspects of a date can be displayed by using TO_CHAR function SELECT Emp_Name "Employee Name", TO_CHAR(Emp_DOB, 'MONTH DD, YYYY') "Date of Birth" FROM Employee
Other datatypes – BINARY_FLOAT, BINARY_DOUBLE, INTEGER, LONG, TIMESTAMP, etc.	